Unique Generalized Abbreviations (hard)

Problem Statement

Given a word, write a function to generate all of its unique generalized abbreviations.

A generalized abbreviation of a word can be generated by replacing each substring of the word with the count of characters in the substring. Take the example of “ab” which has four substrings: “”, “a”, “b”, and “ab”. After replacing these substrings in the actual word by the count of characters, we get all the generalized abbreviations: “ab”, “1b”, “a1”, and “2”.

Note: All contiguous characters should be considered one substring, e.g., we can’t take “a” and “b” as substrings to get “11”; since “a” and “b” are contiguous, we should consider them together as one substring to get an abbreviation “2”.

Example 1:

```
Input: "BAT"
Output: "BAT", "BA1", "B1T", "B2", "1AT", "1A1", "2T", "3"
```

Example 2:

```
Input: "code"
Output: "code", "cod1", "co1e", "co2", "c1de", "c1d1", "c2e", "c3",
"1ode", "1od1", "1o1e", "1o2", "2de", "2d1", "3e", "4"
```

Try it yourself

Try solving this question here:

Java

```java
import java.util.*;

/*class AbbreviatedWord {
  StringBuilder str;
  int start;
  int count;

  public AbbreviatedWord(StringBuilder str, int start, int count) {
    this.str = str;
    this.start = start;
    this.count = count;
  }
}*/

class Solution {

  public List<String> generateGeneralizedAbbreviation(String word) {
    List<String> result = new ArrayList<String>();
    // TODO: Write your code here
    return result;
  }
```

Test Case | Run Code Result | Stdout

"BAT"

Use Example Testcases | Run | Submit

← Previous
Solution: Balanced Parentheses

Next →
Solution: Unique Generalized Abbreviations

Mark as Completed